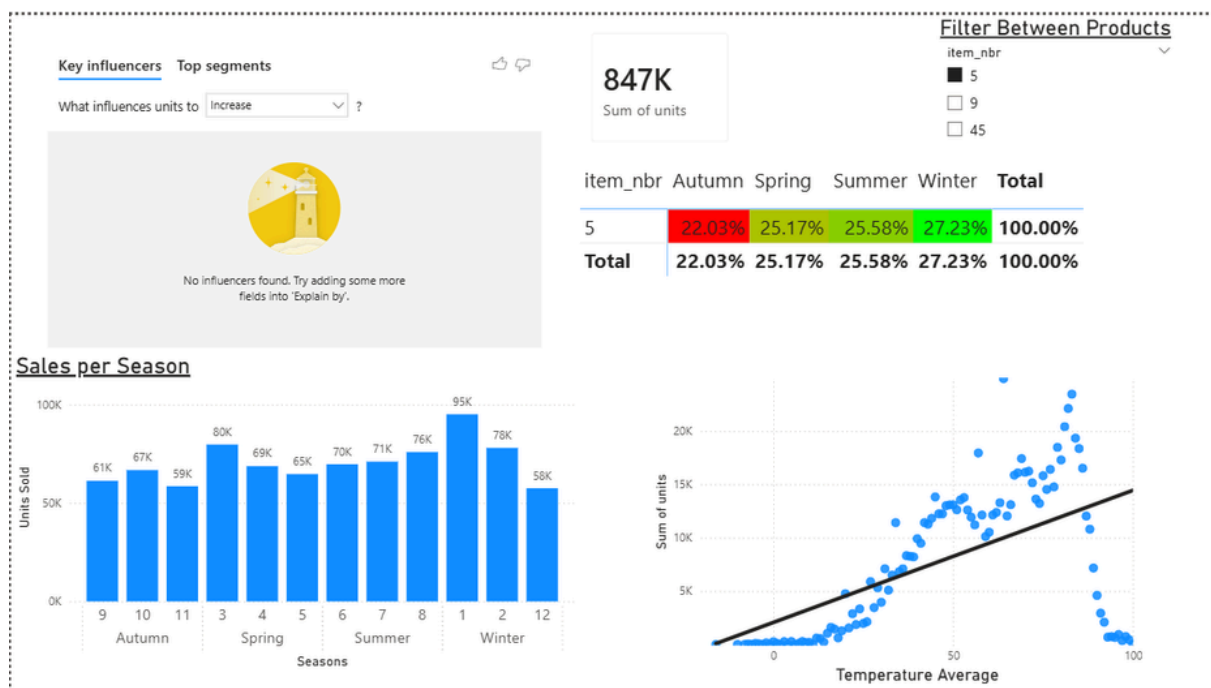


Section C: Interpretation & Visualisation

Product 5– Interpretation, Visualisations and Business Recommendations



For Product 5, my visualisations show that average temperature (tavg) and seasonality have the strongest influence on units sold. Evidence for this is present in my Temperature vs Sales scatter plot. From an overview, we can see that the black trend line shows that there is a positive linear relationship between units sold and temperature average– as temperature increases, so do the units sold. And this makes sense from a business perspective, as people would be more inclined to go into the store and shop for our products in warmer temperatures. However, closer inspection

reveals that a linear model heavily misrepresents the scatter plot's true structure, which is actually curvilinear. In the first section of the scatter plot, the overall slope of the graph is initially a straight line on the x-axis until approximately 18 degrees. After 18 degrees, we see a positive relationship between the average temperature and the sum of units; we have a positive slope with minor fluctuations at 59 degrees and 79 degrees. However, after 83 degrees, we have a sharp decrease in the gradient until sales go back to 0. This inference shows us that high sales for product 5 take place between roughly 50 and 80 degrees. This makes sense because 80 degrees is roughly 27 degrees celcius, after this the temperature starts to become unbearable, which will deter customers from entering our stores to purchase products.

Furthermore, Seasonality has a prominent influence on units sold. Looking at both the heatmap and the clustered bar graph, our highest sales take place in winter(accounting for 27.23% of total sales).

Interestingly, this does partially oppose the pattern our scatter graph was telling us, as in the scatter graph we saw that most sales take place between 50-80 degrees (10-27 degrees in Celsius). However, 10-27 degrees is quite a broad range, and 10 degrees can include normal winter temperatures. Perhaps on the scatter graph between the ranges of 50-80 degrees, we have a higher proportion of sales in the winter phase- this makes sense as there are visibly more blue plots between 50-65 degrees than 65-80 degrees.

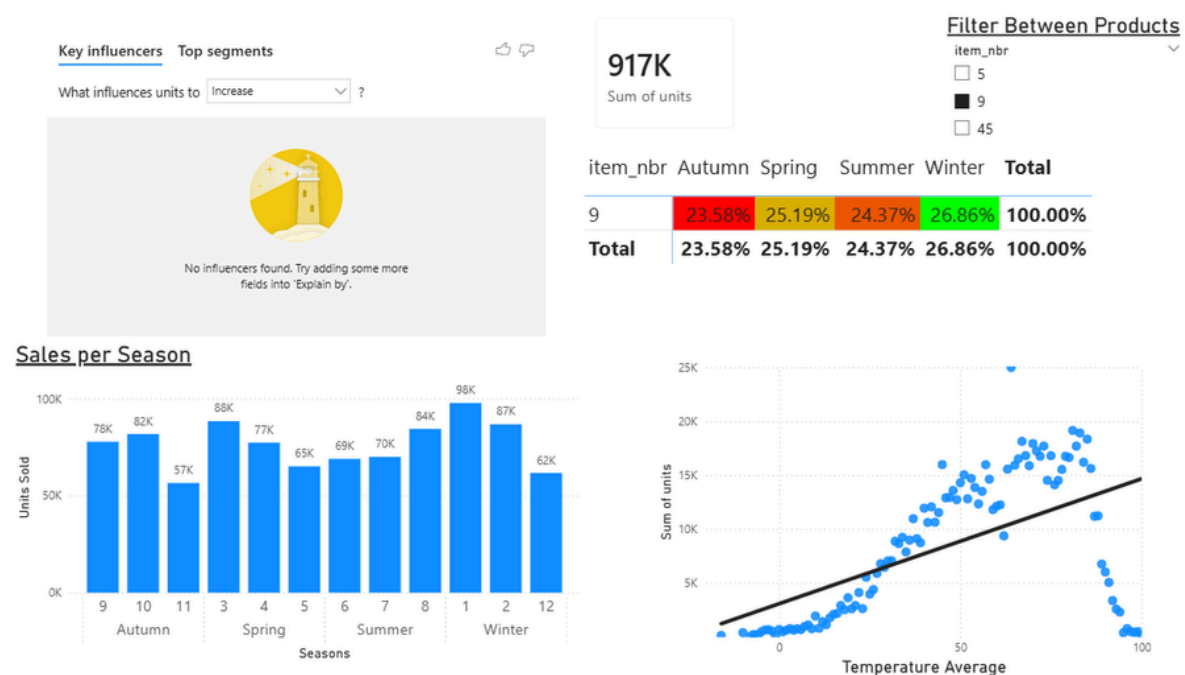
Alternatively, our stores span across various locations, and the regional climate baseline would

differ in different regions; 50 degrees Fahrenheit can be a normal winter day depending on the region. There is a Key Influencer Visual on my report page, but unfortunately, no influencers were found. This is because the Key Influencer visual only shows influencers if it crosses a statistical threshold. Because item number 5 is heavily tied to Winter seasonality and quite curved, the AI could not find a linear factor to display and analyse. Looking at my predictive models, in the summary stats, we saw that our R-squared was a low value of 0.00198. This makes sense now, as our graph shows that we do not have a linear relationship between weather and units sold. We also got an estimated value of -0.14, meaning that when holding all variables constant, rainy days result in a 14% decrease in units sold. This makes sense as fewer people would be inclined to go into our stores when the weather is bad.

From a business perspective, stock for product 5 should be increased in relatively warm temperatures, as our graph shows peaks from 53 to 80 degrees. It should also be increased in the months January-March, as those are the months with the largest sales. To combat lower sales in autumn (as seen in the heatmap), the business should consider implementing autumn-related sales, such as Halloween and Labour Day Sales, to encourage more customers to buy more of product 5. In terms of extreme heat shown in the scatter plot, the business should consider strategies to encourage more customers to go into stores and buy using summer-related incentives, such as buying product 5 comes with a summer product (like a fan) at 50% off. Measures like this

would encourage customers to come into stores when it's extremely warm, as they would be enticed to buy product 5 since it comes with a useful summer item. In conclusion, for product 5, temperature average(tavg) and seasonality have the strongest influence on units sold.

Product 9– Interpretation, Visualisations and Business Recommendations



Similar to product 5, for product 9, my visualisations show that average temperature (tavg) and seasonality have the strongest influence on units sold. This is evident in my Temperature vs Sales scatter plot. From an overview, we can see that the black trend line shows that there is a positive linear relationship between units sold and temperature average– as temperature increases, so do the units sold. And this makes sense from a business perspective, as people would be more

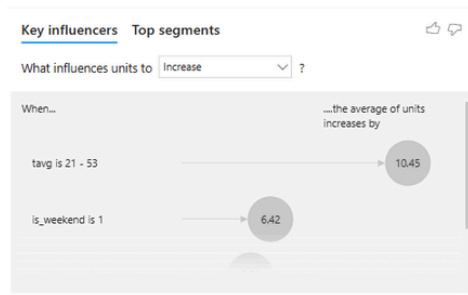
inclined to visit our various stores and purchase products in person when the weather is warm. However, closer inspection reveals that a linear model misrepresents the scatter plot's true structure, which (similarly to product 5) is actually curvilinear. Looking at the first section of the scatter plot, the overall slope is flat and resting on the x-axis. Unlike product 5, product 9 starts to form a slanted slope at 10 degrees and continues on until 80 degrees(with minor fluctuations). After 80 degrees, a similar pattern follows, and sales sharply decrease. After this temperature, the weather starts to become unbearable, which deters customers from entering our stores. Product 9 has its plots grouped closer together (precision) as compared to product 5, hinting that product 9 on a scatter plot is more representative. The scatter plot shows us that high sales for product 9 take place between 45- 80 degrees, a slightly larger scope compared to product 5.

This temperature trend follows along nicely with our heatmap, showing that winter and spring have the highest percentage total of sales. Autumn, however, doesn't follow this trend; this could be because in certain regions, autumn doesn't follow the 45-80 degree range. Or similarly to product 5, there is no incentive to buy our product in autumn (business recommendations on this will be discussed later on). The clustered bar graph shows that our top-selling months are January to March(similar to product 5). Just like with product 5, there is a Key Influencer Visual on my report page, but unfortunately, no influencers were found here. This is because the Key Influencer visual only shows

influencers if it crosses a statistical threshold. Because item number 9 is heavily tied to Winter, the AI could not find a linear factor to display and analyse. Looking at my predictive models, the R-squared value was extremely low, and the shape of our scatter graph justifies this.

From a business perspective, our product does badly in the autumn and excellently in the winter. Similar to product 5, we need to increase our stock in the winter and perhaps slightly increase the price of product 9 in calm temperatures to combat the low sales in autumn. In Autumn, we need to convince consumers to purchase our products by using sales related to public holidays(as people shop more on public holidays). Furthermore, in the extreme heat of the summer, we should try a similar approach to product 5 by tying product 9 with products commonly bought in the summer, such as AC units or Ice cube makers. To conclude, for product 9, temperature average(tavg) and seasonality have the strongest influence on units sold.

Product 45- Interpretation, Visualisations **and Business Recommendations**



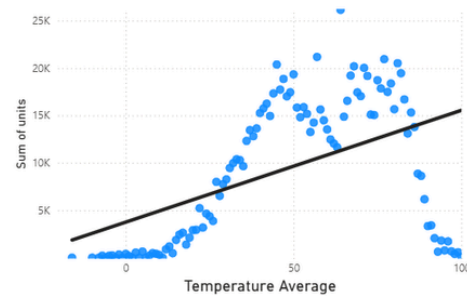
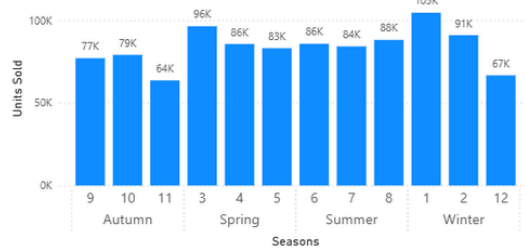
1M
Sum of units

Filter Between Products

item_nbr
☐ 5
☐ 9
☒ 45

item_nbr	Autumn	Spring	Summer	Winter	Total
45	21.86%	26.38%	25.67%	26.09%	100.00%
Total	21.86%	26.38%	25.67%	26.09%	100.00%

Sales per Season



Product 45 is our best-selling product, accounting for 33% of total sales(1M). Overall average temperature holds the strongest influence for product 45, and this is justified in multiple visualisations. To start, despite our trend line being a straight, slanted slope, the scatter graph follows a curvilinear structure. Our gradient starts to increase between 13- 80 degrees, compared to our other 2 products, this is the widest range of increase in sales. However, between 50 and 80 degrees, our data starts to become messier and lose its positive relationship(particularly between 65-80 degrees). Past 80 degrees, the sales start to decrease as the temperature becomes too hot to enjoy shopping. The majority of our sales take place during a normal temperature range (50-80 degrees), which makes sense from a business perspective, temperatures between this range are optimal for going out and spending money(multiplier effect).

The heatmap produces some interesting insights, the

colour is green for Spring, Summer and Winter but red for Autumn. The colour was red for Autumn in products 5 and 9, but here there is a larger disparity between Autumn and the rest of the months; even our top-selling product cannot beat the low sales of Autumn. It is clear now that this is not a product issue but a seasonal issue. The clustered bar graph for Sales per Season also displays this trend, with 2 of our 3 lowest-selling months being in Autumn (November and September). Key influencers work well with our added features from the table cleaned in R. It states that when `tavg` is between 21–53 degrees, the average of units increases by 10.45. This links back to my main point from the scatter graph: our highest sales take place in optimal temperatures. Furthermore, when `is_weekend` is 1 (the day is Saturday or Sunday), the average of units increases by 6.42. This links back to the point I made when initially creating the feature, `is_weekend` on R (I briefly mentioned this in a comment). This point is also corroborated by my linear regression model, the estimate for `is_weekend` was 0.1. When holding all variables constant, the weekends see a 10% increase in units sold compared to weekdays. This makes sense as people would have more free time on the weekends and therefore can purchase more items in the store. Additionally, when setting the key influencers to 'decrease' for units, it concluded that when `tavg` is 61–64, the average of units decreases by 10.6. This is a large factor and does oppose our scatter graphs, 61–64 degrees Fahrenheit converts to 16–18 degrees, which is inside our optimal temperature range. Quite fascinating, this localised dip could represent a transitional climate threshold where the behaviour of

our consumer temporarily shifts.

Business recommendations are quite apparent here; we have our lowest sales for the top 3 products in autumn, so we must convince customers to visit our store in autumn. We can do this by creating product bundles where we partner some of our top-selling products alongside a top-selling autumn product. This is partnered with tailoring our store towards the autumn season with decorations and special autumn discounts tailored to our top 3 products. We can also implement social media challenges to boost our popularity in the autumn. This is all because our top 3 products all suffer a sales loss in autumn, meaning that our stores may not be desirable to customers during this time period. Implementing these strategies can provide meaningful increases in units sold. To conclude, average temperature holds the strongest influence for product 45.